

Understanding Pro-Environmental Behaviour Among the Youth of India

Evidence from a Survey of 35 Students Living in Delhi — A Data Analysis and Reproducible Research Project

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Introduction & Rationale

Environmental degradation — poor air quality, climate change, waste accumulation, and resource depletion — is one of the defining challenges of this century. Young people are widely regarded as the most environmentally aware demographic, but whether that concern translates into concrete pro-environmental action remains a central puzzle in behavioural research.

Why Delhi?

One of the world's most air-polluted major cities (annual mean PM2.5 far above WHO limits) and a national education hub attracting students from many states — ideal for testing whether pollution environment relates to concern and action.

Why This Matters

If concern does not automatically become action, spending on awareness alone is inefficient. Identifying the exact lever that converts intent into behaviour helps target limited resources where they have the most impact — a directly policy-usable result.

Theoretical Framework

The study is grounded in Ajzen's (1991) **Theory of Planned Behavior (TPB)**, the dominant model for explaining and predicting environmentally relevant behaviour, and Stern's (2000) **Value-Belief-Norm theory**, which links concern to behaviour through values and awareness of consequences. In this report, **concern** aggregates knowledge, concern, and perceived urgency; **action** aggregates five self-reported behaviours.

Construct	Definition	Measured As
Attitude	Evaluative judgement of the behaviour (good or bad, worthwhile or not)	'Acting environmentally is important.'
Subjective Norm	Social pressure from important others to perform the behaviour	'People who matter to me expect it.'
Perceived Behavioural Control	Perceived ease or difficulty of performing the behaviour (barriers and resources)	'It is easy for me to make green choices.'

- ❏ A widely replicated finding in this literature is the **intention–behaviour gap**: people state strong environmental intentions yet underperform them. This paper quantifies that gap among Delhi youth.

Objectives & Hypotheses

Research Objectives

- Measure environmental concern, intention, and self-reported action among Delhi-based youth.
- Relate behavioural scores to the pollution environment of each student's home state.
- Compare Delhi natives with interstate migrant students on concern, intention, and action.

Hypotheses & Outcomes

H1 — Supported ✓

Higher environmental concern leads to more pro-environmental action (positive).

H2 — Not Supported ✕

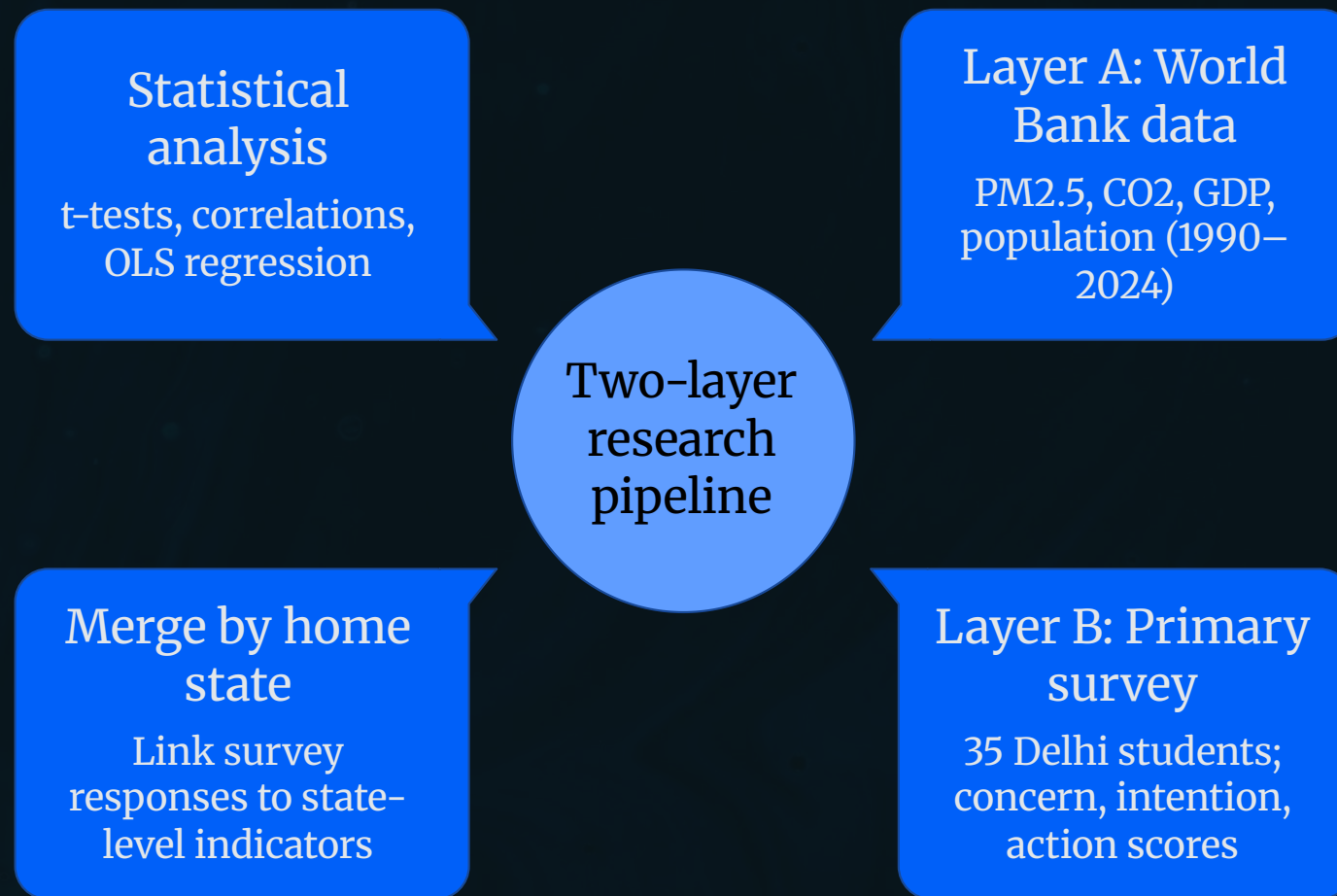
Living in a more polluted home environment leads to higher environmental concern.

H3 — Not Supported ✕

Migrants from highly polluted states show higher concern than Delhi natives.

Methodology: Two-Layer Design

The project uses a two-layer design. **Layer A** supplies objective environmental context from open-source data. **Layer B** supplies the behavioural response from a primary survey. Merging them by home state yields a per-respondent pollution context that powers the analysis.



Layer A pulls India's time series (1990–2024) via the World Bank Open Data API. Layer B is a purpose-designed TPB survey collecting demographics, concern items, intention items, and five self-reported behaviours. Scores are computed for concern (1–5), intention (1–5), action (0–5), and the intention–behaviour gap (intention minus action).

Survey Instrument & Sample Profile

Survey Structure

01

Demographics

Age, gender, city, and state of origin.

02

Concern

Awareness, concern, urgency — 1 to 5 scale.

03

Intention (TPB)

Attitude, subjective norm, perceived control, intention — 1 to 5 scale.

04

Self-Reported Behaviour

Waste segregation, reusable bag, green transport, e-waste recycling, participation — Yes / Sometimes / No.

05

Open-Ended

What makes acting green hard or easy — free text.

Sample Profile

35

Total Respondents

20/15

Natives / Migrants

15

States Represented

19–25

Age Range (yrs)

Gender was mixed (M and F). Migrants covered 15 states including Uttar Pradesh, Bihar, Punjab, Rajasthan, Kerala, Tamil Nadu, Maharashtra, and others.

Key Results: Concern Predicts Action

Pearson Correlations

Relationship	r	p-value
Concern ↔ Action	0.335	0.049 ✓
Intention ↔ Action	-0.097	0.579
Home PM2.5 ↔ Concern	0.104	0.554
Home PM2.5 ↔ Action	0.108	0.536

The relationship between concern and action is **positive and statistically significant** ($r = 0.335$, $p = 0.049$) — the headline finding. Intention alone does not significantly predict action, confirming it is a weaker lever than concern in this sample.

OLS Regression

$$\text{Action} = \beta_0 + \beta_1(\text{Concern}) + \beta_2(\text{Intention}) + \varepsilon$$

β_0 Intercept

0.710

β_1 Concern

+0.646

β_2 Intention

-0.008

R^2

0.112

Concern is the dominant positive driver; intention contributes almost nothing. The two predictors explain about **11% of variance** in action — modest but meaningful for self-reported behaviour.

The Intention–Behaviour Gap & Group Comparisons

3.8/5

Average Concern

High environmental concern across the full sample.

+0.48

Gap: Natives

Delhi natives intend to act more than they report doing.

+1.08

Gap: Migrants

Migrants show a larger intention–behaviour gap.

r=0.34

Concern→Action

Statistically significant correlation (p = 0.049).

Welch’s t-Test: Native vs. Migrant

Variable	t	p	Significant?
Concern	1.383	0.179	No
Intention	-0.549	0.587	No
Action	1.540	0.137	No
Gap	-1.611	0.120	No

All four tests are non-significant (p > 0.05). The shared experience of living in Delhi dominates home-state differences, pointing toward **city-level rather than state-level environmental socialisation**.

India’s Environmental Trend (Layer A)

India’s ambient PM2.5 exposure and per-capita CO₂ emissions were pulled live from the World Bank Open Data API (1990–2024). Home-state PM2.5 showed only a weak, non-significant relation to concern (r = 0.10) and action (r = 0.11), confirming H2 is not supported.

Discussion & Limitations

Key Interpretations

- **Concern drives action** ($r = 0.34$, $p = 0.049$, coefficient $+0.65$) fostering genuine concern is an effective route to behaviour change.
- **The gap is the actionable finding:** students already intend to act but don't. The barrier is implementation (perceived behavioural control), not motivation.
- **No native–migrant difference** (all $p > 0.05$) suggests Delhi's shared environment visible pollution, campus culture, information environment neutralises state-of-origin differences.

Limitations

- Small, non-random sample ($n = 35$) — illustrative, not population-generalisable.
- Self-report bias may inflate the intention–behaviour gap.
- Home-state PM2.5 values are indicative annual means, not station-level measurements.
- Survey data are synthetic (demonstration) and must be replaced with real collected responses.
- Correlation $\neq$ causation; cross-sectional design cannot establish temporal ordering.
- Single-city focus limits generalisation to other Indian cities or rural youth.

Conclusion & Recommendations

Young people in Delhi are highly concerned about the environment, but this concern is not fully converted into everyday pro-environmental action. **Concern is a genuine and statistically significant predictor of action, whereas intention is not.** The persistent intention–behaviour gap reveals that the binding constraint is practical rather than motivational.



Waste Infrastructure

Accessible waste-segregation bins and convenient e-waste collection points on campuses and in public spaces.



Green Transport

Affordable and incentivised public and active transport options to lower the barrier to sustainable commuting.



Campus Green Defaults

Institutional defaults that make sustainable behaviour the easy choice targeting the perceived behavioural control lever identified by TPB.



Reproducible Pipeline

The full project is regenerable in two commands: fresh open data pull, recomputed statistics, and redrawn figures — transparent and verifiable by anyone.

❏ **Core Recommendation:** Shift resources from awareness-raising toward removing practical barriers. Policies that lower implementation barriers are likely to yield more behaviour change than further awareness campaigns.